

ZOMATO RESTAURANT DATA ANALYSIS USING PYTHON

DATASET OVERVIEW

DESCRIPTION OF THE DATASET

The dataset contains information about restaurants listed on Zomato. The dataset consists of twenty-two columns, which are **Sl No, Restaurant ID, Restaurant Name, Country Code, City, Address, Locality, Locality Verbose, Longitude, Latitude, Cuisines, Average Cost for two, Currency, Has Table booking, Has Online delivery, Is delivering now, Switch to order menu, Price range, Aggregate rating, Rating color, Rating text, and Votes**.

NUMBER OF ROWS AND COLUMNS

The dataset contains **9,552 rows** and **22 columns**.

DATASET OVERVIEW

MISSING VALUES IDENTIFIED

The columns containing missing values are Locality (1), Locality Verbose (1), Longitude (1), Latitude (1), Cuisines (10), Average Cost for two (2), Currency (1), Has Table booking (1), Has Online delivery (1), Is delivering now (1), Switch to order menu (1), Price range (2), Aggregate rating (2), Rating color (2), Rating text (2), and Votes (2).

DATA TYPE CHANGES PERFORMED

The data cleaning process involved **converting** the 'Average Cost for two' column to a numeric type, **filling missing values** in key numerical columns (Average Cost for two, Price range, Aggregate rating, and Votes) with their **means**, and **removing** misaligned **rows** based on the CSV structure.

DUPLICATE RECORDS (IF ANY)

No duplicate records were found in the dataset.

DATA ANALYSIS

STATISTICAL SUMMARY

...	Country Code	Average Cost for two	Price range	Aggregate rating	Votes
count	9550.000000	9550.000000	9550.000000	9550.000000	9550.000000
mean	18.348272	1199.326387	1.804607	2.666314	156.923037
std	56.728197	16122.023220	0.905378	1.516447	430.189709
min	1.000000	0.000000	1.000000	0.000000	0.000000
25%	1.000000	250.000000	1.000000	2.500000	5.000000
50%	1.000000	400.000000	2.000000	3.200000	31.000000
75%	1.000000	700.000000	2.000000	3.700000	131.000000
max	216.000000	800000.000000	4.000000	4.900000	10934.000000

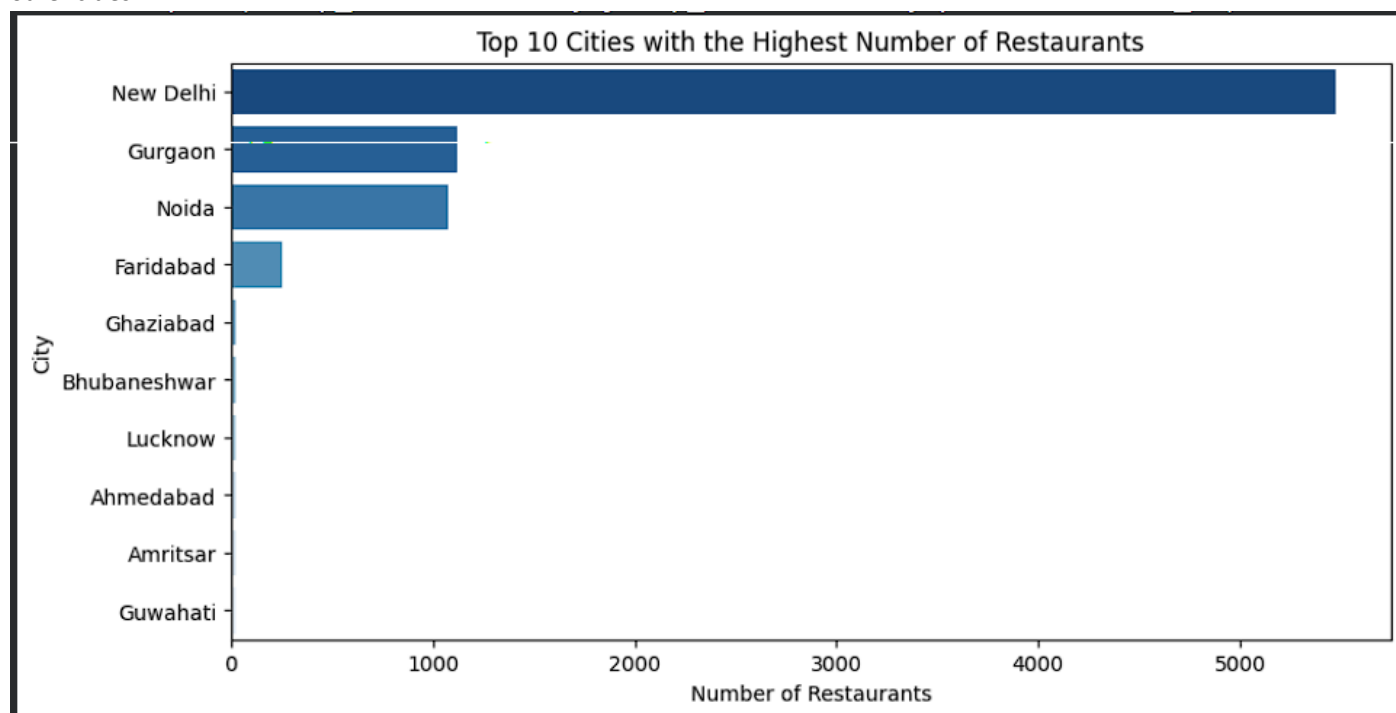
ANSWERS TO THE ANALYSIS QUESTIONS AND KEY OBSERVATIONS

1. What is the average restaurant rating?
The average restaurant rating is 2.67.
2. Which restaurant has the highest number of votes?
The restaurant with the highest number of votes is **Toit** with 10934.0 votes.
3. What is the average cost for two across all restaurants?
The average cost for two across all restaurants is 1199.33.
4. How do the average ratings compare between restaurants that have online delivery and those that do not?
Restaurants with online delivery have an average rating of 3.25, while those without online delivery have an average rating of 2.47. **Restaurants with online delivery generally have higher ratings.**
5. How do the average ratings compare between restaurants that have table booking and those that do not?
Restaurants with table booking have an average rating of 3.44, while those without table booking have an average rating of 2.56. **Restaurants offering table booking generally receive better ratings.**
6. What are the top 10 cities with the highest number of restaurants?
The top 10 cities with the highest number of restaurants are New Delhi (5,473), Gurgaon (1,118), Noida (1,080), Faridabad (251), Ghaziabad (25), Ahmedabad (21), Bhubaneswar (21), Lucknow (21), Guwahati (21), and Amritsar (21).
7. Which city has the highest restaurant presence?
New Delhi has the highest restaurant presence with 5473 restaurants.
8. What is the most popular cuisine?
The most frequently appearing cuisine combination is **North Indian**.
9. Do restaurants with online delivery generally have higher ratings?
Yes, **restaurants with online delivery generally have higher ratings** (average 3.25) compared to those without (average 2.47).
10. Do restaurants offering table booking receive better ratings?
Yes, **restaurants offering table booking generally receive better ratings** (average 3.44) compared to those that do not (average 2.56).
11. What did you learn from analyzing this dataset?
Data analysis is an effective tool to interpret business insights.

DATA VISUALIZATIONS

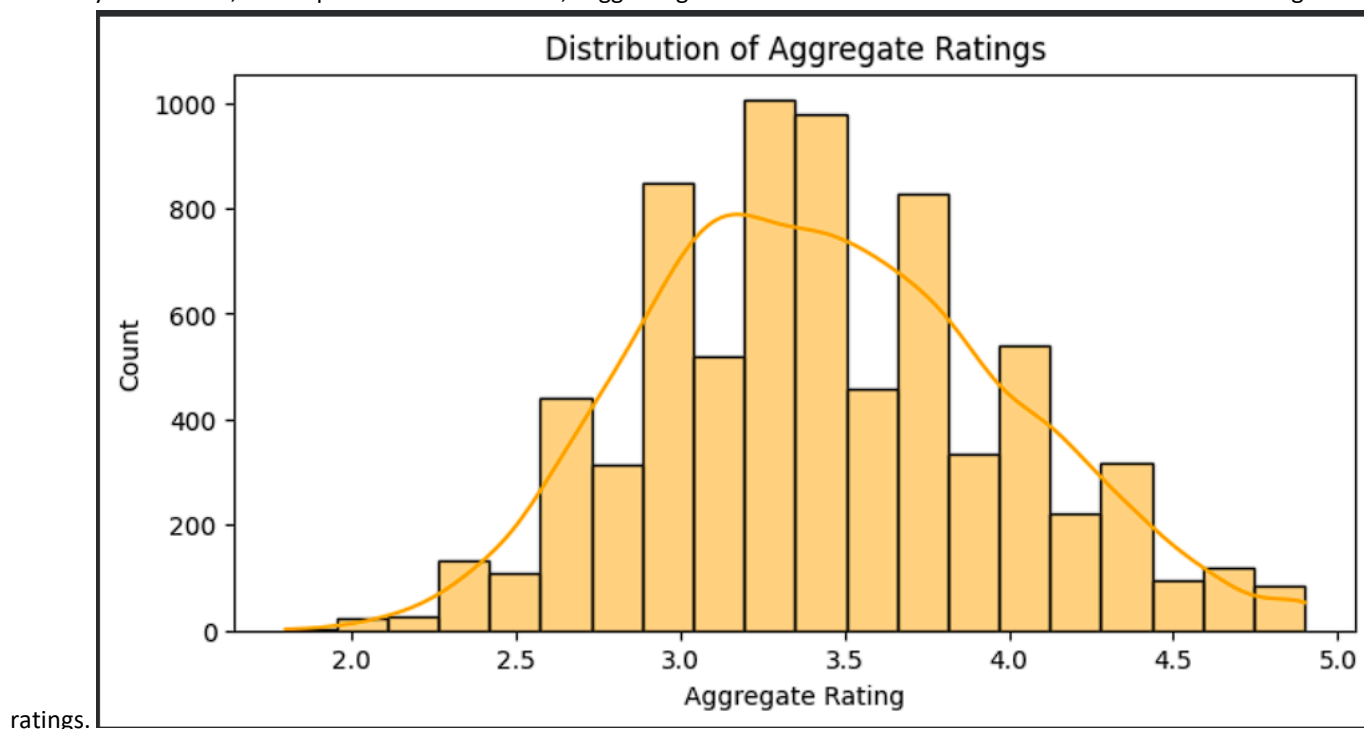
TOP 10 CITIES WITH THE HIGHEST NUMBER OF RESTAURANTS

This horizontal bar plot displays the top 10 cities that have the highest number of restaurants listed in the dataset. It clearly shows that New Delhi significantly dominates in restaurant presence, followed by Gurgaon and Noida, with a steep drop-off for other cities.



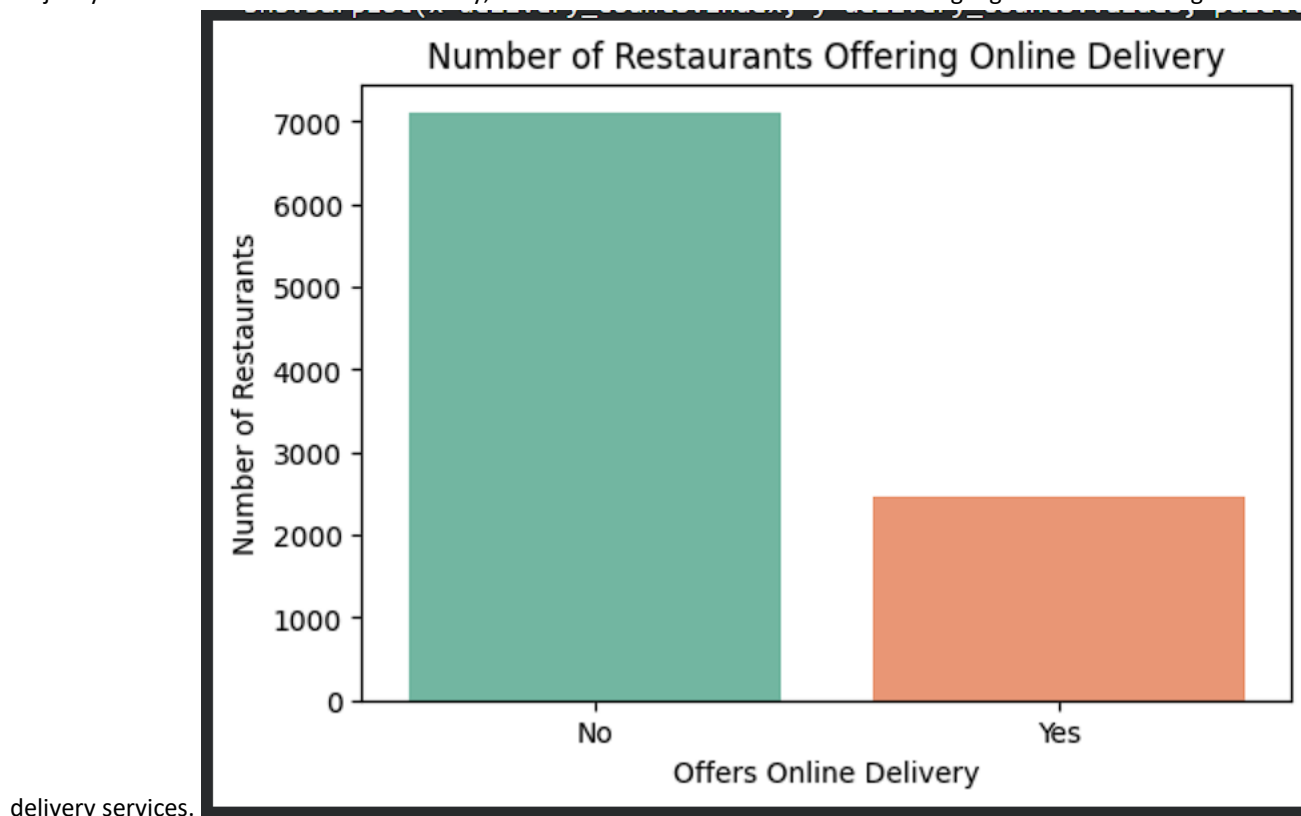
DISTRIBUTION OF AGGREGATE RATINGS

This histogram visualizes the frequency distribution of restaurant aggregate ratings. It indicates that the ratings are roughly normally distributed, with a peak around 3.0 to 3.5, suggesting that most restaurants in the dataset have moderate to good ratings.



NUMBER OF RESTAURANTS OFFERING ONLINE DELIVERY

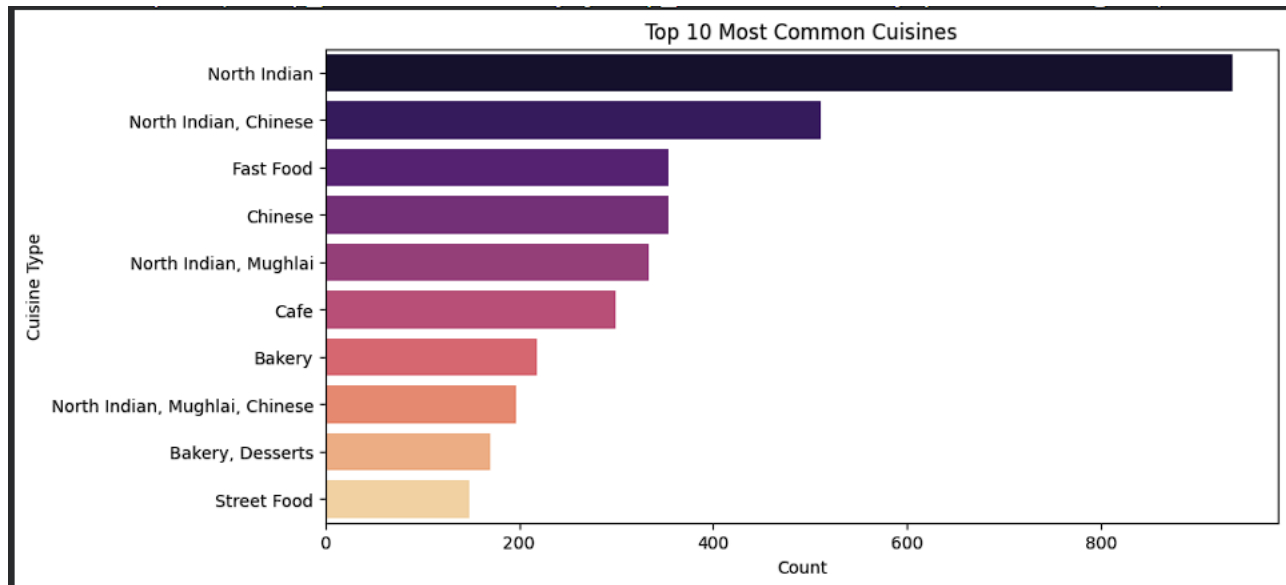
This bar chart illustrates the count of restaurants that offer online delivery versus those that do not. The plot shows a clear majority of restaurants offer online delivery, while a substantial number do not. This highlights the market segment for online



TOP 10 MOST COMMON CUISINES

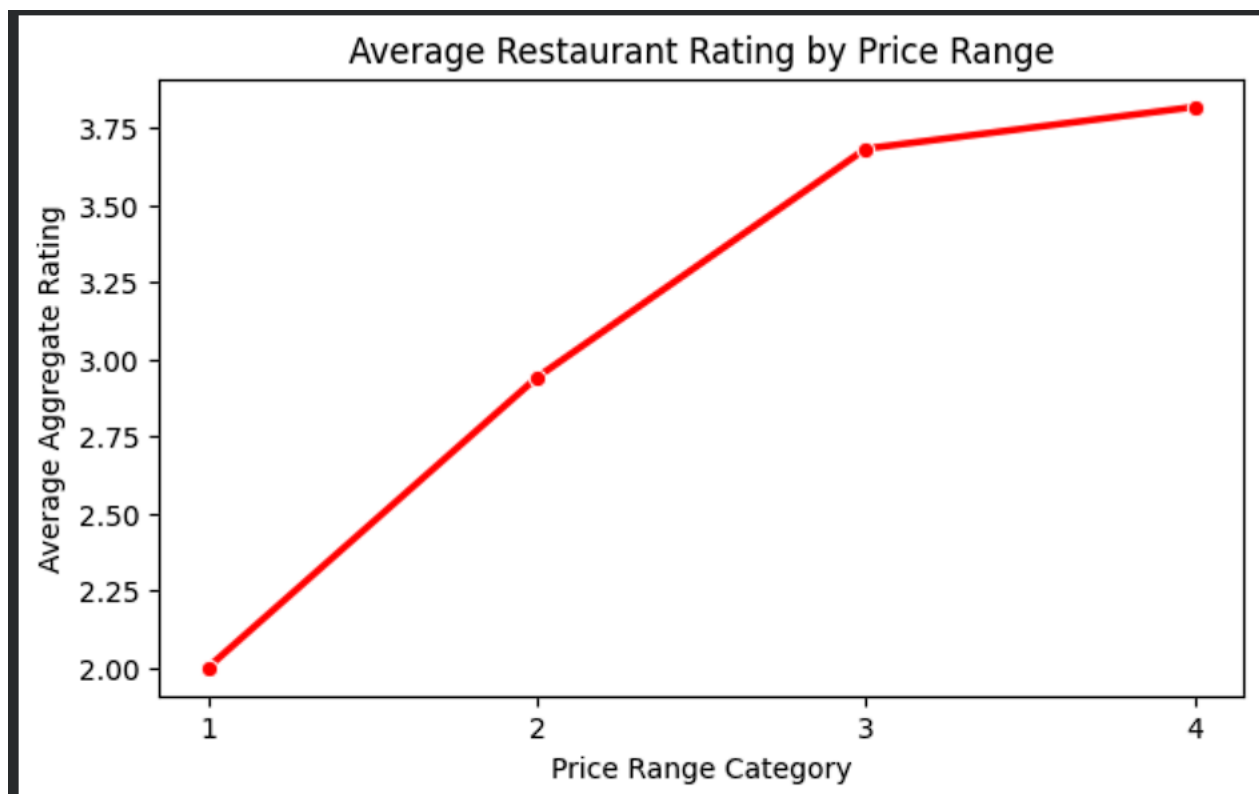
This horizontal bar chart presents the top 10 most frequently appearing cuisine types or combinations in the dataset. It reveals that 'North Indian' is by far the most common cuisine, followed by 'North Indian, Chinese', and other popular options like 'Fast

Food' and 'Chinese', indicating regional preferences and dining trends.



AVERAGE RESTAURANT RATING BY PRICE RANGE

This line plot shows how the average aggregate rating of restaurants changes across different price ranges. The plot demonstrates a positive correlation: as the price range category increases (from 1 to 4), the average aggregate rating also tends to increase, suggesting that more expensive restaurants generally receive higher ratings.



CONCLUSION

- New Delhi has a significantly higher number of restaurants, indicating a competitive market. Other cities might offer better growth opportunities.
- North Indian cuisine, alone or combined with Chinese, is the most common, showing strong consumer preference.
- Restaurants offering online delivery generally have higher average ratings (3.25) compared to those that don't (2.47).
- Restaurants with table booking facilities also tend to have better average ratings (3.44) than those without (2.56).
- Most restaurants have moderate to good ratings (around 3.0-3.5), suggesting a market with decent but not always exceptional dining experiences.

- Higher-priced restaurants tend to receive higher average ratings, indicating that customers often associate higher cost with better quality.
- Restaurants can improve customer satisfaction and ratings by offering online delivery and table booking services.
- Factors like convenience and service (beyond just food quality) are crucial for overall customer satisfaction and good ratings.